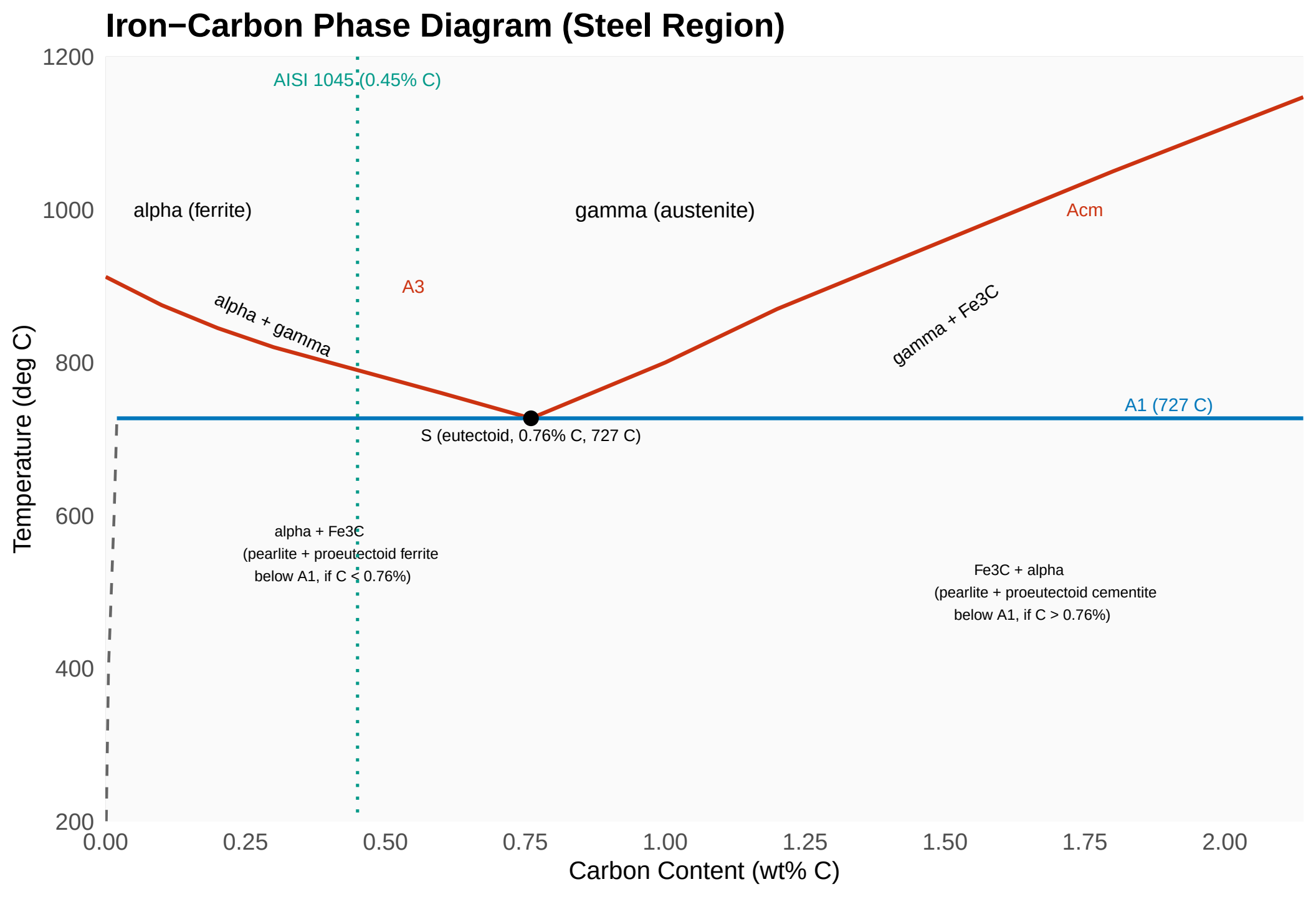
Iron–Carbon Phase Diagram (Steel Region)
AISI 1045 (0.45% C)
alpha (ferrite)
gamma (austenite)
Acm
A3
alpha + gamma
gamma + Fe3C
A1 (727 C)
S (eutectoid, 0.76% C, 727 C)
alpha + Fe3C
(pearlite + proeutectoid ferrite
below A1, if C < 0.76%)
Fe3C + alpha
(pearlite + proeutectoid cementite
below A1, if C > 0.76%)
Temperature (deg C)
Carbon Content (wt% C)
1200
1000
800
600
400
200
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00